

Benjamin Choon Heng Lee

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🔗 Google Scholar

RESEARCH EXPERIENCE

JPMorgan Chase & Co.

New York, NY, USA

Senior Associate Research Scientist at *Global Technology Applied Research*

June 2024 - Present

- I engage and work closely with firm stakeholders to understand existing business needs and practices, identify potential use cases with immersive technologies, and conduct scholarly research to evaluate suitability and inform product decisions
- I co-design and develop an open-source 3D data visualization toolkit used by other academic researchers and by engineers within the firm
- I deliver live demonstrations, research talks, and educational seminars at firmwide events of up to 1000+ attendees and to internal and external C-suite executives
- I publish research in academic venues including ACM CHI and IEEE VIS
- Manager: Dr. Blair MacIntyre

University of Stuttgart

Stuttgart, Baden-Württemberg, Germany

Postdoctoral Researcher at *Visualization Research Center (VISUS)*

February 2023 - April 2024

- I mentored and closely supervised four doctoral students on ongoing projects, refining research questions and methods, which led to top-tier publications and one successful thesis defence (remainder in progress)
- I led independent research projects with lab mentors and external collaborators
- Managers: Prof. Michael Sedlmair & Prof. Dieter Schmalstieg

Microsoft Research

Redmond, WA, USA

Research Intern at *Visualization and Interactive Data Analysis Group*

June 2019 - September 2019

- I conceptualized a novel approach for communicating data to general audiences by eliciting visceral emotional responses in virtual reality
- Mentors: Dr. Dave Brown, Dr. Steven Drucker, Prof. Bongshin Lee

Monash University

Melbourne, Victoria, Australia

Doctoral Researcher at *Data Visualization and Immersive Analytics Lab*

February 2019 - January 2023

- I conducted research in the combined use of flat surfaces with the virtual 3D space enabled by immersive head-mounted devices for the purposes of interactive data visualisation and immersive analytics
- Advisors: Prof. Tim Dwyer, A/Prof. Bernhard Jenny, A/Prof. Maxime Cordeil, Dr. Arnaud Prouzeau

EDUCATION

Monash University

Melbourne, Victoria, Australia

Ph.D. in Immersive Analytics

February 2019 - January 2023

- Thesis: *Surfaces and Spaces in Immersive Analytics*
- Advisors: Prof. Tim Dwyer, A/Prof. Bernhard Jenny, A/Prof. Maxime Cordeil, Dr. Arnaud Prouzeau
- Examiners: Prof. Harald Reiterer, Prof. Luciana Nedel

Monash University

Melbourne, Victoria, Australia

Bachelor of Informatics and Computation Advanced (First Class Honours)

2015 - 2018

- Thesis: *Heterogeneous Mixed-Reality Display Environments for Immersive Visual Analytics*
- Advisors: Prof. Tim Dwyer, A/Prof. Bernhard Jenny, A/Prof. Maxime Cordeil

PUBLICATIONS

The following lists exclude short papers, posters, workshops, and extended abstracts. Please see my Google Scholar page for a complete list.

Journal Articles

* Denotes equal contribution

- 2026 [Hybrid User Interfaces: Past, Present, and Future of Complementary Cross-Device Interaction in Mixed Reality](#)
Sebastian Hubenschmid*, Marc Satkowski*, Johannes Zagermann*, Julián Méndez*, Niklas Elmqvist, Steven Feiner, Tiare Feuchtner, Jens Emil Grønbæk, **Benjamin Lee**, Dieter Schmalstieg, Raimund Dachsel, and Harald Reiterer
Under review at IEEE Transactions on Visualization and Computer Graphics
doi: 10.48550/ARXIV.2509.05491
- 2026 [Design Considerations for Visualization Transitions of 3D Spatial Data in Hybrid AR-Desktop Environments](#)
Yucheng Lu, Tobias Rau, **Benjamin Lee**, Andreas Köhn, Michael Sedlmair, Christian Sandor, and Tobias Isenberg
To appear in Computer Graphics Forum
- 2026 [Situating Brushing and Linking in Virtual and Augmented Reality](#)
Carlos Quijano-Chavez, **Benjamin Lee**, Nina Doerr, Wolfgang Büschel, Michael Sedlmair, and Dieter Schmalstieg
To appear in IEEE Transactions on Visualization and Computer Graphics (IEEE VR 2026)
- 2025 [Desktop versus VR for collaborative sensemaking](#)
Ying Yang, Tim Dwyer, Zachari Swiecki, **Benjamin Lee**, Michael Wybrow, Maxime Cordeil, Teresa Wulandari, Bruce H. Thomas, and Mark Billinghurst
Frontiers in Virtual Reality
doi: 10.3389/frvir.2025.1570383
- 2024 [Visual Highlighting for Situated Brushing and Linking](#)
Nina Doerr, **Benjamin Lee**, Katarina Baricova, Dieter Schmalstieg, and Michael Sedlmair
Computer Graphics Forum (EuroVis 2024)
doi: 10.1111/cgf.15105
- 2024 [Design Patterns for Situated Visualization in Augmented Reality](#)
Benjamin Lee, Michael Sedlmair, and Dieter Schmalstieg
IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2023)
doi: 10.1109/TVCG.2023.3327398
- 2024 [PerSiVal: On-Body AR Visualization of Biomechanical Arm Simulations](#)
Xingyao Yu, David Rosin, Johannes Kässinger, **Benjamin Lee**, Frank Dürr, Christian Becker, Oliver Röhrle, and Michael Sedlmair
IEEE Computer Graphics and Applications (invited talk at IEEE VR 2026)
doi: 10.1109/MCG.2024.3494598
- 2024 [Lights, Headset, Tablet, Action: Exploring the Use of Hybrid User Interfaces for Immersive Situated Analytics](#)
Xiaoyan Zhou*, **Benjamin Lee***, Francisco R. Ortega, Anil Ufuk Batmaz, and Yalong Yang
Proceedings of the ACM on Human-Computer Interaction (ACM ISS 2024)
doi: 10.1145/3698147
- 2022 [Towards immersive collaborative sensemaking](#)
Ying Yang, Tim Dwyer, Michael Wybrow, **Benjamin Lee**, Maxime Cordeil, Mark Billinghurst, and Bruce H. Thomas
Proceedings of the ACM on Human-Computer Interaction (ACM ISS 2022)
doi: 10.1145/3567741
- 2021 [Data Visceralization: Enabling Deeper Understanding of Data Using Virtual Reality](#)
Benjamin Lee, Dave Brown, Bongshin Lee, Christophe Hurter, Steven Drucker, and Tim Dwyer

IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2020)
doi: 10.1109/TVCG.2020.3030435

🏆 **Honorable Mention Award (Top 5% of Submissions)**

- 2021 [Shared Surfaces and Spaces: Collaborative Data Visualisation in a Co-located Immersive Environment](#)
Benjamin Lee, Xiaoyun Hu, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer
IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2020)
doi: 10.1109/TVCG.2020.3030450

- 2021 [The MADE-Axis: A Modular Actuated Device to Embody the Axis of a Data Dimension](#)
Jim Smiley, **Benjamin Lee**, Siddhant Tandon, Maxime Cordeil, Lonni Besançon, Jarrod Knibbe, Bernhard Jenny, and Tim Dwyer
Proceedings of the ACM on Human-Computer Interaction (ACM ISS 2021)
doi: 10.1145/3488546

🏆 **Honorable Mention Award (Top 5% of Submissions)**

Conference Proceedings

* Denotes equal contribution

- 2026 [S-TIER: Situated-Traceable Insights in Extended Reality for Hybrid Crisis Management](#)
Feiyu Lu*, Cheng Yao Wang*, Leonardo Pavanatto, Fannie Liu, David Saffo, **Benjamin Lee**, Mengyu Chen, and Blair MacIntyre
Under submission to IEEE International Symposium on Mixed and Augmented Reality
- 2026 [ARCurrencyCrittters: an Augmented Reality Game for Supporting Historically Marginalized Youth's Financial Literacy](#)
Erica Principe Cruz, David Saffo, Leonardo Pavanatto, **Benjamin Lee**, Feiyu Lu, Youngwook Do, Mengyu Chen, Cheng Yao Wang, Blair MacIntyre, and Fannie Liu
Under review at ACM Interaction Design and Children Conference
- 2026 [Anu.js: Accelerating Web-based Immersive Analytics](#)
David Saffo*, **Benjamin Lee***, Feiyu Lu, Cheng Yao Wang, and Blair MacIntyre
To appear in ACM CHI Conference on Human Factors in Computing Systems
- 2025 [Examining the Effects of Immersive and Non-Immersive Presenter Modalities on Engagement and Social Interaction in Co-located Augmented Presentations](#)
Matt Gottsacker, Mengyu Chen, David Saffo, Feiyu Lu, **Benjamin Lee**, and Blair MacIntyre
ACM CHI Conference on Human Factors in Computing Systems
doi: 10.1145/3706598.3713346
- 2025 [Traversing Dual Realities: Investigating Techniques for Transitioning 3D Objects between Desktop and Augmented Reality Environments](#)
Tobias Rau, Tobias Isenberg, Andreas Koehn, Michael Sedlmair, and **Benjamin Lee**
ACM CHI Conference on Human Factors in Computing Systems
doi: 10.1145/3706598.3713949
- 🏆 **Best Paper Award (Top 1% of Submissions)**
- 2024 [Design Space of Visual Feedforward And Corrective Feedback in XR-Based Motion Guidance Systems](#)
Xingyao Yu, **Benjamin Lee**, and Michael Sedlmair
ACM CHI Conference on Human Factors in Computing Systems
doi: 10.1145/3613904.3642143
- 2023 [Deimos: A Grammar of Dynamic Embodied Immersive Visualisation Morphs and Transitions](#)
Benjamin Lee, Arvind Satyanarayan, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer
ACM CHI Conference on Human Factors in Computing Systems
doi: 10.1145/3544548.3580754
- 2022 [A Design Space For Data Visualisation Transformations Between 2D And 3D In Mixed-Reality Environments](#)
Benjamin Lee, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer
ACM CHI Conference on Human Factors in Computing Systems
doi: 10.1145/3491102.3501859
- 🏆 **Honorable Mention Award (Top 5% of Submissions)**

2022 [Tangible Globes for Data Visualisation in Augmented Reality](#)
 Kadek Ananta Satriadi, Jim Smiley, Barrett Ens, Maxime Cordeil, Tobias Czauderna, **Benjamin Lee**,
 Ying Yang, Tim Dwyer, and Bernhard Jenny
ACM CHI Conference on Human Factors in Computing Systems
 doi: 10.1145/3491102.3517715

AWARDS

- **Best Paper Award (Top 1% of Submissions)** 2025
ACM CHI Conference on Human Factors in Computing Systems
- **Vice-Chancellor's Commendation For Thesis Excellence Award** 2023
Monash University
- **Honorable Mention Award (Top 5% of Submissions)** 2022
ACM CHI Conference on Human Factors in Computing Systems
- **Honorable Mention Award (Top 5% of Submissions)** 2021
ACM International Conference on Interactive Surfaces and Spaces
- **Honorable Mention Award (Top 5% of Submissions)** 2021
IEEE Visualization Conference

ACADEMIC SERVICE

- **Program Committee** 2026
ACM CHI Conference on Human Factors in Computing Systems
- **Program Committee** 2025, 2026
ACM International Conference on Interactive Surfaces and Spaces
- **Program Committee** 2025
IEEE Visualization Conference
- **Student Volunteer** 2022
IEEE Conference on Virtual Reality and 3D User Interfaces
- **Student Volunteer** 2020
Australian Conference on Human-Computer Interaction

REVIEWING

- CHI, ISS 2026
- CHI, ISMAR, ISS, VIS, VR · TVCG 2025
- CHI, EuroVis, ISMAR, ISS, MobileHCI, SIGGRAPH, VIS, VR, VRST · IJHCI, TVCG 2024
- ISMAR, ISS, UIST, VIS, VR, VRST · Frontiers, IJHCI, JCSS, TVCG 2023
- CHI, ISMAR, MobileHCI, VIS, VR 2022
- CHI, EuroVis, ISMAR, ISS, VIS 2021
- VIS 2020
- CHI 2019

SUPERVISION

Doctoral Students

- Co-advisor of **Tobias Rau** Since 2024
Topic: AR for Computational Chemistry
With Prof. Michael Sedlmair
- Co-advisor of **Nina Doerr** Since 2023
Topic: Visual Highlighting in the Real World
With Prof. Michael Sedlmair and Prof. Dieter Schmalstieg
- Co-advisor of **Carlos Quijano-Chavez** Since 2023
Topic: Interaction Techniques for Situated Analytics
With Prof. Dieter Schmalstieg and Prof. Michael Sedlmair
- Co-advisor of **Xingyao Yu** 2023 – 2025
Thesis Title: On-Body Visualization with Extended Reality: Exploration and Application
With Prof. Michael Sedlmair

Master's Students

- Co-advisor of **Xiaoyan Hu** 2019
Thesis Title: Collaborative Data Visualisation in Virtual Reality
With Prof. Tim Dwyer

Bachelor's Students

- Advisor of **Vivien Schraitle** 2023 – 2024
Thesis Title: Supporting Cross-reality Transfer of Virtual Objects in Hybrid Desktop and Augmented Reality Environments

TEACHING

- **Head Teaching Assistant** at Monash University Semester 2 2021 & Semester 1 2022
FTI5147 - Data Visualisation and Exploration
- **Teaching Assistant** at Monash University Semester 2 2020
FIT3146 - Maker Lab
- **Teaching Assistant** at Monash University Semester 1 2020 & Semester 1 2021
FTI5147 - Data Visualisation and Exploration

INVITED TALKS

- "Web-based Immersive Analytics with Anu.js" Spring 2025
New York University
- "Immersive Analytics: Data Visualization and Exploration using Immersive Technologies" Fall 2024
New York University
- "Immersive and Situated Analytics" June 2023
University of Stuttgart
- "Immersive and Situated Analytics" May 2023
Graz University of Technology